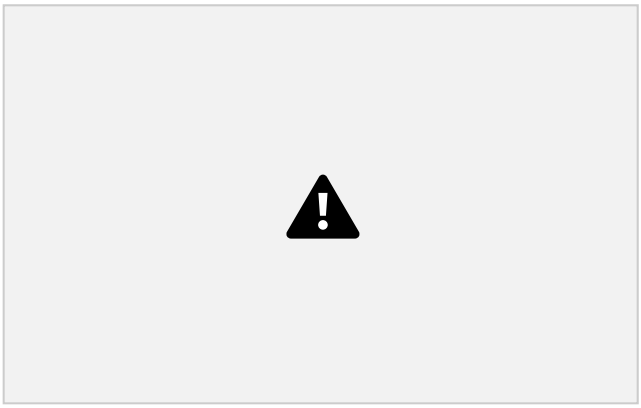
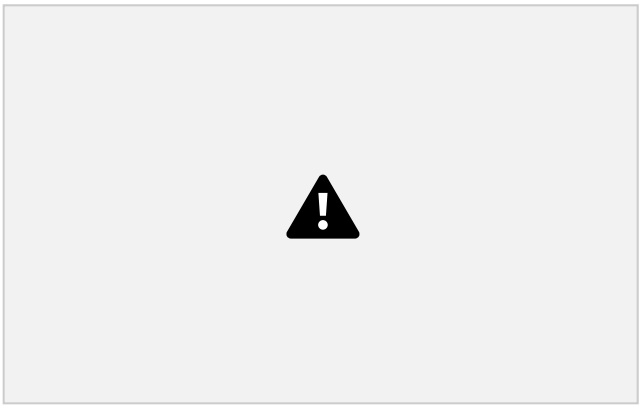
DSA06 - Data Handling and Visualization | Activity: Identify Skewness using Violin Plots | Slot D

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Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

Chart 1: Exam marks (marks)Chart 2: Commute distance (km)

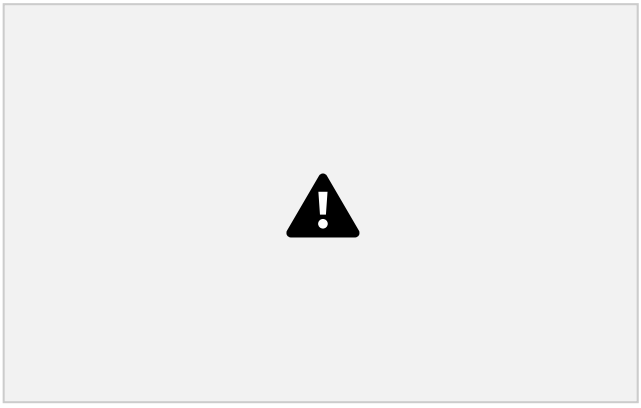




Violin	Skew (circle one)	Mean vs Median (circle)
Class B	Right / Left / Symm.	Mean > / < / ~ Median
Class C	Right / Left / Symm.	Mean > / < / ~ Median
Class A	Right / Left / Symm.	Mean > / < / ~ Median

Violin	Skew (circle one)	Mean vs Median (circle)
Dept IT	Right / Left / Symm.	Mean > / < / ~ Median
Dept AIDS	Right / Left / Symm.	Mean > / < / ~ Median
Dept CSE	Right / Left / Symm.	Mean > / < / ~ Median

Chart 3: Monthly income (Rs thousand)Chart 4: Fuel efficiency (km per litre)





Violin	Skew (circle one)	Mean vs Median (circle)
Suburb	Right / Left / Symm.	Mean > / < / ~ Median
Metro	Right / Left / Symm.	Mean > / < / ~ Median
Town	Right / Left / Symm.	Mean > / < / ~ Median

Part B (2 marks)

Violin	Skew (circle one)	Mean vs Median (circle)
Brand X	Right / Left / Symm.	Mean > / < / ~ Median
Brand V	Right / Left / Symm.	Mean > / < / ~ Median
Brand W	Right / Left / Symm.	Mean > / < / ~ Median

B1. In Chart 2, violin "Dept IT" is not symmetric. Towards which side is its tail, and which is the better measure of a typical commute distance here: mean or median? Give one reason. [1]

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1].

Part A: How to identify the answers

Skewness	Mean vs. Median
Right-skewed	Mean > Median
Left-skewed	Mean < Median
Symmetric	Mean ≈ Median

Use this rule for all groups in Charts 1–4.

Part B

B1. Chart 2 – Dept IT (1 mark)

The exact answer depends on the shape of the Dept IT violin plot.

- **Tail towards higher commute distances:** Right-skewed; the **median** is better because extreme values can pull the mean upwards.
- **Tail towards lower commute distances:** Left-skewed; the **median** is better because extreme values can pull the mean downwards.

B2. ggplot2 code (1 mark)

```
ggplot(df, aes(x = group, y = value)) +
  geom_violin() +
  stat_summary(fun = median, geom = "point", colour = "red")
```

Explanation: `geom_violin()` draws the violin plots, and `stat_summary()` displays the median as a red point.

Please upload the image containing Charts 1–4, and I can fill in every row with the exact answers.